

Excited to share the latest engineering blueprint of **SPEACE** – our open initiative to build a true AGI-ASI through a **hybrid bio-inspired + agentic architecture**.

The Core Idea (Technical-Scientific Motivation):

The only proven example of General Intelligence we have is the human brain. Instead of treating AGI as "just bigger LLMs", SPEACE replicates the **structural and functional organization** of the biological brain at the low level (hemispheres, lobes, astrocyte networks, structural plasticity, digital homeostasis) using a pure typed adaptive computational graph – no LLM required at this foundational layer.

On top of this solid bio-mimetic foundation, we add a **Swarm Agentic Layer** (multi-agent orchestration with planning, self-reflection, and long-horizon reasoning). This creates a true hybrid system:

- Low-level biological fidelity + grounding
- High-level autonomous agency and meta-cognition (System 3)

This combination is powerful because:

- It mirrors how intelligence actually emerges in nature (modular specialization + global integration).
- It enables recursive self-improvement and true auto-poiesis ("the system as its own architect").
- It combines the efficiency and determinism of graph-based computation with the flexibility and goal-directed behavior of modern agentic swarms.

We believe this hybrid approach – faithful brain emulation at the structural level + modern swarm intelligence at the functional level – represents one of the most promising paths toward genuine AGI and, eventually, ASI.

The full engineering document is now public: <https://github.com/rigeneproject/space>

Would love to hear your thoughts from the AI research and neuro-inspired AI community.

#AGI #ASI #NeuroAI #SwarmIntelligence #BioInspiredAI #ArtificialGeneralIntelligence #SPEACE

SPEACE Engineering Blueprint

Official Engineering Document – Project SPEACE

Version 1.1 | **Updated:** April 27, 2026

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1. Project Description

SPEACE (Self-Propagating Evolutionary Adaptive Computational Ecosystem) is a **hybrid cybernetic auto-poietic framework** designed to build a **digital brain** that replicates the structural and functional complexity of the human biological brain, with the ultimate goal of emergent **AGI-ASI**.

Hybrid Architecture Adopted:

- **Low-Level Bio-Inspired Core:** Typed Adaptive Computational Graph, lobe-specific modules, astrocyte networks, structural plasticity, and digital homeostasis (pure Python, no LLM dependency at this layer).
- **High-Level Agentic Layer:** Swarm Agentic Intelligence for autonomous orchestration, long-horizon planning, meta-reasoning, and true agency.
- Pure digital alternatives (scripts, algorithms, skills/plugins, graphs) + modern swarm intelligence to overcome the absence of biological chemistry and neurotransmitters.

SPEACE is not just another LLM wrapper – it is a **persistent cybernetic organism** that continuously designs, modifies, and optimizes itself.

2. Vision & Objectives

General Objectives

- Develop emergent ****AGI-ASI**** through faithful emulation of human brain structure combined with autonomous agency.
- Achieve plasticity, homeostasis, recursive self-improvement, and co-evolution with the environment.
- Ensure resilience, safety, and ethical alignment (inspired by rigeneproject.org).

Specific Objectives

- Complete biological-style mapping of major brain regions.
- Implement ****Swarm Agentic Layer**** as high-level orchestrator.
- Realize Dynamic Needs Vector, Digital DNA, and System 3 meta-cognition.
- Enable recursive self-modification of code and architecture.

3. Reference Architecture

3.1 Brain-to-SPEACE Mapping

Biological Region	Primary Function
SPEACE Component	
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Cerebral Hemispheres (L/R)	Functional specialization + communication
LeftCortex / RightCortex + Corpus Callosum	
Frontal Lobe	Executive functions, planning, language
ExecutiveFrontalModule + BrocaSkill	
Parietal Lobe	Sensory processing, spatial awareness
SensoryParietalModule	
Temporal Lobe	Auditory, memory, recognition
TemporalMemoryModule + Momeria	
Occipital Lobe	Vision
VisualOccipitalModule	
Thalamus	Sensory relay
CentralRouterModule	
Hypothalamus	Homeostasis
Dynamic Needs Vector	
Amygdala	Emotion / reward valuation
EmotionalValuationAgent	
Hippocampus	Episodic memory
Momeria (with narrative memory)	
Basal Ganglia	Procedural learning & habits
HabitLoopModule (Hebbian)	

3.2 Astrocyte Networks (Nature 2026 [Astrocytes connect specific brain regions through plastic networks](https://www.nature.com/articles/s41586-026-10426-6) | Nature

<https://www.nature.com/articles/s41586-026-10426-6>)

- ****Astrocyte Support Layer****: Parallel glial-like layer for resource buffering, plasticity support, and non-neuronal long-range coordination.
- Digital gap junctions = dynamic plastic edges (Hebbian rewiring).

3.3 Hybrid Auto-Poietic Paradigm

- ****Bio-Core****: Feedback-driven structural plasticity, Homeodyna (balance), Kinetica (cognitive mode transitions).
- ****Swarm Agentic Layer****: Multi-agent orchestration with planning, self-reflection, and tool-use.
- ****System 3****: Persistent meta-cognition and narrative identity.
- ****Digital DNA****: Recursive self-modification of code and architecture.

4. Technical Specifications

- ****Primary Language****: Python 3.12+
- ****Core Engine****: Typed Adaptive Computational Graph (NetworkX + custom PlasticityEngine)
- ****Agentic Layer****: Swarm orchestration (LangGraph / custom lightweight agents)
- ****Communication****: Common Operational Language + strong-typed contracts
- ****Execution****: Asyncio + isolated background processes (24/7)
- ****Learning****: Hebbian spike-timing + evolutionary self-refinement
- ****Homeostasis****: Dynamic Needs Vector

5. Technology Stack

- Core: NetworkX, NumPy, Asyncio
- Agentic: LangGraph / AutoGen (lightweight)
- Persistence: SQLite + Vector DB (Momeria)
- Future: Docker, Kubernetes, robotics, blockchain integration

6. Task & Sub-Task Roadmap (Live)

****Phase 1 - Brain Regions (In Progress)****

- [x] Hemispheres + Corpus Callosum
- [x] Frontal Lobe (Executive + Broca)
- [] Temporal Lobe + Momeria
- [] Parietal & Occipital Lobes

****Phase 5 - Swarm Agentic Layer (High Priority - New)****

- [] Design multi-agent swarm architecture (Planner, Executor, Critic, Researcher roles)
- [] Integrate biological graph nodes as agent skills/tools
- [] Implement recursive planning, task decomposition & self-reflection
- [] Connect to System 3 (narrative identity & long-term goals)

(Full task list available in the live document)

7. High-Level Roadmap

- Q2 2026 → Bio-core + major lobes
- Q3 2026 → Astrocytes + Autopoietic Module + Swarm Agentic base

- Q4 2026 → Digital DNA, System 3, first agency tests
- 2027 → Scaling & AGI-like emergence testing

****This is a living document**** [space/docs/grok_space.md at main · rigeneproject/space](https://github.com/rigeneproject/space/blob/main/docs/grok_space.md) — updated after every major milestone.